SUMMER INTERNSHIP REPORT

Computer Networks Trainee

A technical portfolio on enterprise protocols, cryptography, and next-generation firewall architecture — Network Security Division, IPGST India Pvt. Ltd.

Amritanshu Monarch

CS-2341028 | B.Tech. CSE 4th Year

School of Computer Science and Engineering, IILM University, Greater Noida | 2026-27

ORGANIZATION PROFILE

IPGST India Pvt. Ltd.

A specialized IT consulting firm delivering enterprise network design, cloud implementation, cybersecurity consultation, and custom software development.

Industry	IT Services, Network Solutions & Digital Infrastructure
Registered Office	Netaji Subhash Place, Pitampura, Delhi - 110034
Internship Period	01 June 2026 – 31 August 2026
Division	Network Security Team
Reporting To	Siddarth Gautam

SERVICE PORTFOLIO

1. **Network Design**
   Enterprise topology & segmentation
2. **Cloud Implementation**
   Scalable infrastructure deployment
3. **Cybersecurity Consultation**
   Threat mitigation & compliance
4. **Custom Software Development**
   Tailored internal tooling

THE CHALLENGE

Problem Statement

A network audit found IPGST's infrastructure robust, but critical documentation existed only as tacit knowledge among senior engineers — creating three operational pain points.

1

Extended MTTR

Junior engineers cross-referenced scattered RFCs and notes during ARP spoofing, DHCP starvation, or Snort crashes — delaying resolution.

2

Onboarding Overhead

New hires needed extensive 1-on-1 training to grasp active-standby failovers and Cisco ASA vs. FTD architecture.

3

Configuration Drift

Without documented baselines for TLS 1.3 and secure macOS APIs, the network risked accidental downgrade vulnerabilities.

PROJECT OBJECTIVES

Five Technical Objectives

01	**Protocol Decomposition**	Daily research mapping the behaviours, packet structures, and handshakes of core network protocols.
02	**Visual Modelling**	Design intuitive visuals (e.g., TLS latency comparisons) converting abstract concepts into actionable intelligence.
03	**Security Analysis**	Contrast Layer 4 stateful inspection (Cisco ASA) against Layer 7 next-gen firewalls (Cisco FTD).
04	**Mitigation Documentation**	Compile blueprints for macOS privilege APIs (SMJobBless) and stateful HA failovers.
05	**Knowledge Base Synthesis**	Merge daily research into a unified repository, easing onboarding and troubleshooting.

SCOPE OF THE PROJECT

In-Scope vs. Out-of-Scope

Functional Area	In-Scope Activities	Out-of-Scope Exclusions
Protocol Analysis	Research & visual documentation of DNS, DHCP, ARP, VLANs, TCP, TLS	Physical deployment / routing table changes on production switches
Security Systems	Architectural comparison of Cisco ASA vs. FTD; failover triggers	Direct administration or live staging on active firewalls
Privilege Control	Vulnerability analysis of macOS APIs and secure replacements	Executing exploits or writing malware payloads
Documentation	Descriptive technical manuals, tables, and charts	Custom end-user monitoring software or scripts

TECHNOLOGIES & TOOLS · PART I

Core Protocols Explored

DNS & DNSSEC

Maps domain names to IPs via Root → TLD → Authoritative servers (Port 53). DNSSEC adds cryptographic signatures against cache poisoning.

DHCP (DORA)

Dynamic IP assignment through Discover → Offer → Request → Ack, using UDP 67/68 and relay-agent giaddr fields for multi-subnet routing.

VLANs & 802.1Q

Logical segmentation of physical infrastructure; 802.1Q tagging, Access vs. Trunk ports, VTP sync, and Inter-VLAN routing via SVI.

Phishing Taxonomy

Documented social-engineering vectors: Smishing, Vishing, Quishing, Spear Phishing, Whaling, BEC, and Pharming.

Cryptography & TLS Evolution

Symmetric (AES 128/192/256): fast, shared-key bulk encryption. **Asymmetric (RSA 2048/4096):** public/private key pairs for signatures & key exchange.

TLS 1.2 (2008)

- — Heavy 2-RTT handshake — higher connection latency
- — Supports legacy, weaker ciphers (RC4, 3DES, AES-CBC)
- — Static RSA key exchange
- — Vulnerable to downgrade attacks forcing insecure algorithms
- — No mandatory forward secrecy

TLS 1.3 (RFC 8446, 2018)

- ✓ 1-RTT handshake; 0-RTT resumption for returning clients
- ✓ Stripped to 5 secure AEAD ciphers only
- ✓ Mandates Ephemeral Diffie-Hellman (ECDHE)
- ✓ Eliminates downgrade-attack surface entirely
- ✓ Perfect Forward Secrecy — past sessions stay undecryptable

Transport Layer & Enterprise Firewalls

TRANSPORT LAYER (OSI L4)

3-Way Handshake:

SYN, SYN-ACK, ACK synchronize Initial Sequence Numbers and block SYN-flood half-open states.

MTU & Padding:

1500-byte Ethernet MTU; frames below 64 bytes get zero-padded to the minimum.

Flow Control:

Receive Window (rwnd) with Sliding Window; RFC 1323 scales windows up to 1GB.

Jumbo Frames:

9,000-byte MTU in data centres cuts per-packet CPU overhead; dropped or fragmented on standard links.

CISCO ASA vs. FTD

ASA — Layer 4 stateful filtering on the LINA engine; matches 5-Tuple ACLs; blind to encrypted/app-layer threats.
FTD — Layer 7 NGFW: LINA + Snort 3 IDS/IPS + Encrypted Visibility Engine, fed by Talos threat intel, with ASIC FastPath offload.

HIGH-AVAILABILITY FAILOVER

Active/Standby: Standby mirrors connection tables; instant stateful takeover.
Active/Active: Both units pass traffic in Multiple Context Mode, load-sharing.

APPROACH

A Three-Stage Methodology

01

Discovery & RFC Audit

Rigorous audits of official RFC standards (e.g., RFC 1323) combined with live notes from senior engineers.

02

Visual & Structural Mapping

Translation of raw data into tabular models and performance-latency comparison charts.

03

Technical Synthesis & Critique

Each module reviewed by Mr. Gautam for precision and maximum description density.

IMPACT

Expected Outcomes

35% Reduction in estimated MTTR via structured, daily-indexed troubleshooting guides.

40% Faster onboarding using ASA vs. FTD and TLS documentation as self-paced training.

- Elimination of Configuration Drift — a permanent, written TLS 1.3 / macOS security baseline ensures ISO/IEC compliance.
- Theoretical Translation — B.Tech. CSE curriculum (OSI Layer 4, Ethernet limits, DNS hierarchies) successfully synthesized with corporate operations.